Gut–pulmonary vascular–right ventricular interactions in PAH

A Intestinal changes



Intestine (schematic)



Small-intestinal mucosa (separate schematic)

Gut microbiota

Barrier impairment

Microbial compositional / functional shifts

Candidate ecological response

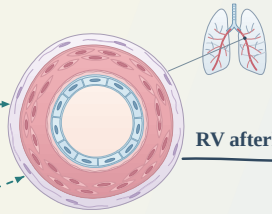
Gut alterations

B Pulmonary vasculature

MAMP translocation / exposure

Altered metabolite exposure

Microbial function + host processing

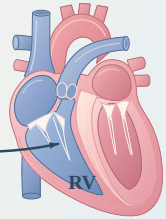


Vascular remodeling

PVR ↑

RV afterload ↑

C Right ventricle



Stage dependent

Adaptation / maladaptive remodeling

With RV dysfunction

Systemic hemodynamic compromise

Reverse links: human heart failure evidence [68–70]

PAH feedback remains proposed; the complete loop is unvalidated.

Venous congestion

Right atrial pressure ↑

Mesenteric hypoperfusion

Reduced mucosal perfusion

Gut wall edema

***L. rhamnosus* GG**

Experimental PH
MCT rats [71]

RV function improved

RV fibrosis reduced

Pulmonary vascular load / pulmonary arteriolar remodeling

No significant differences detected

Microbiota-related signals

Established afterload relation

Hemodynamic effects

Proposed PAH link / incomplete pathway